Mohammad Khalayleh

Software Developer | FPGA & Hardware Verification | Computer Engineer

Jerusalem, Israel | Open to Remote (CET / EST overlap) | +972 52 738 8786 | Khalayleh.eng@gmail.com
Portfolio: mohammed-khalayleh.web.app | GitHub: github.com/mohammedkhalayleh | LinkedIn: linkedin.com/in/mohammad-khalayleh

SUMMARY

Computer Engineering graduate with two years of web development experience building and contributing to real-world applications using React.js, JavaScript, Firebase, Node.js, and related web technologies. Experienced in developing responsive user interfaces, business systems, and multi-user applications. Also gained hands-on hardware experience through an FPGA-based graduation project implementing AES-128 CTR encryption and SHA-256 hashing using SystemVerilog on an Intel Cyclone V FPGA. Interested in software engineering, FPGA development, and hardware verification roles.

TECHNICAL SKILLS

Software Development: JavaScript, React.js, Next.js, Node.js, Express.js, Java, C, Flutter, HTML5, CSS3, Bootstrap, Python.
Hardware & Verification: SystemVerilog, VHDL, RTL Design, FPGA Design, Digital Logic, Finite State Machines, Testbench Development.
Databases & Cloud: Firebase, Firestore, MongoDB, Oracle Database, SQL, Firebase Hosting.
Systems & Networking: Linux, Windows Server, VMware ESXi, Active Directory, Group Policy, TCP/IP, DNS, DHCP, VLANs, ACLs, Routing, FortiGate.
Tools: Git, GitHub, Quartus Prime, ModelSim, Postman.

SELECTED PROJECTS

FPGA-Based Cryptographic System — Graduation Project

SystemVerilog, FPGA, Quartus Prime, ModelSim, Intel Cyclone V

- Designed, implemented, and verified hardware modules for **AES-128 CTR encryption** and **SHA-256 hashing** on a DE10-Standard FPGA board.
- Developed RTL modules, finite-state-machine control, key-expansion and message-padding logic, and self-checking testbenches; reduced AES processing from **36 to 27 clock cycles per block**.

SOOQNA — Project Manager & Full-Stack Developer

React.js, Node.js, Express.js, MongoDB, PayPal, Coinbase, CoinPayments.

- Led the development of a full-stack e-commerce marketplace supporting separate **customer and merchant workflows**, product listings, shopping, order management, and invoice generation.
- Contributed to most of the frontend, backend, database, and integration work; customized an existing React.js template and developed REST APIs using Node.js and Express.js.
- Integrated multiple payment methods, including credit cards, PayPal, and cryptocurrency payments through Coinbase and CoinPayments.

TMS - Trip Management System

React.js, JavaScript, Firebase, React Router, Bootstrap, EmailJS, jsPDF

- Contributed to a production multi-user platform for managing complete travel packages, suppliers, hotels, transportation, guides, and quotations.
- Implemented responsive workflows for trip planning, PDF quotation generation, Firebase data management, and sending requests to multiple suppliers.

EXPERIENCE

Web Developer — Math Academy

2022-2024

- Contributed to the development and maintenance of **Mass Bill**, an Arabic ERP web application for managing finance, accounting, transactions, and business assets.
- Developed responsive and reusable user-interface components using **React.js, JavaScript, Bootstrap, HTML, and CSS**.
- Implemented and tested workflows related to financial records, transactions, accounting operations, and asset management.

International Academic Experience — Mälardalen University, Sweden

Jan 2025 – Jun 2025

- Completed advanced coursework in **Programming of Reliable Embedded Systems** , **Cybersecurity Operations and Artificial Intelligence**.
- Gained international academic experience working in a multicultural environment and adapting to different engineering and teaching approaches.

EDUCATION & CERTIFICATIONS

B.Sc. Computer Engineering — Al-Quds University, Abu Dis | Graduated Jun 2026
Certifications: Responsive Web Design — freeCodeCamp (2026) | Computer Hardware Basics — Cisco Networking Academy (2024) | Full-Stack Web Development— MathAcademy, (2022) | IEEEXtreme 18.0 Programming Competition (2024) | Quality Assurance (QA) – SigmaLabsJRS & Analiza (2024).
Languages: Arabic (Native) · English (Professional Working Proficiency) · Hebrew (Professional Working Proficiency)